Sahaj Bhadja

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Summary

Cloud-focused software engineer with 2+ years building **AWS/Azure production workloads**, Kubernetes and Docker deployments, CI/CD automation, distributed data pipelines, and LLM services; delivered 99%+ reliability and measurable cost savings.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant AI platform on Azure SQL and Blob Storage** for 500+ users, optimizing indexed queries and access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built **ETL-style cloud data pipelines** across FastAPI, SQL Server, PostgreSQL, and storage services, processing 5,000+ monthly jobs with reconciliation, error handling, and CI/CD-backed releases achieving 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented a **production AWS backend** serving 150+ instructors and 10,000+ daily requests, integrating multimodal LLM workflows while maintaining stable APIs and reducing feedback turnaround time by 60%.
- Designed **concurrent batch processing** with multithreading and secure file ingestion, processing 1,000+ files per batch while logging model tokens and costs across fault-tolerant execution paths and CI/CD releases.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **reproducible PyTorch training and evaluation pipelines** with Hydra and Weights & Biases, processing 100K+ trajectory records while improving long-horizon planning 18% across autonomous-driving simulation workloads at scale.
- Architected a **spatial-temporal ML encoder** integrating HD-map geometry and trajectory context through multi-head attention, improving closed-loop planning stability 15% across repeatable model evaluations and rigorous simulation rollouts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Redesigned **AWS Glue and S3 ETL workflows** for JSON, CSV, and Parquet data, improving pipeline efficiency 68%, response times 30%, and storage costs 15% through scalable transformations.
- Automated **cross-platform builds and 20+ test suites** with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, reducing technical debt 20% while strengthening regression validation and release reliability.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Packaged **eight Node.js microservices as Docker artifacts** and deployed them through Jenkins to Azure App Services using ARM templates, cutting release time 50% across staged environments reliably.
- Implemented **Sentry error tracking, Azure Monitor metrics, Kibana logs**, and automated test gates across a platform synchronizing 2M+ records, improving operational visibility and 95%+ build validation rates.

Skills

Cloud: AWS (EC2, S3, RDS, Glue, ECS, CodePipeline), Microsoft Azure, GCP, Docker, Kubernetes
IaC & Automation: Terraform, Ansible, Jenkins, GitHub Actions, CI/CD, Bash, Shell, Git
AI/ML: LLM Systems, OpenAI API, Transformers, PyTorch, TensorFlow, NLP, Feature Engineering, Hydra, W&B
Observability: Sentry, Azure Monitor, Kibana, logging, metrics, error tracking, debugging, performance optimization
Data & Development: Python, SQL, PostgreSQL, MySQL, MongoDB, FastAPI, Flask, REST APIs, ETL, Distributed Systems

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python data-processing API** on AWS ECS/Fargate with CodePipeline, automated validation checks, and CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.

Big-Brain Live | *GCP, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built a **GCP-backed LLM document platform** with FAISS, Pinecone, PostgreSQL, and REST APIs, indexing 50K+ records through incremental updates while reducing retrieval latency 40% for scalable semantic search.

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **Kubernetes-deployed fintech platform** supporting 50,000+ users and 1M+ transactions, combining secure bank linking, real-time updates, MongoDB-backed access, and CI/CD automation for scalable cloud operations.